

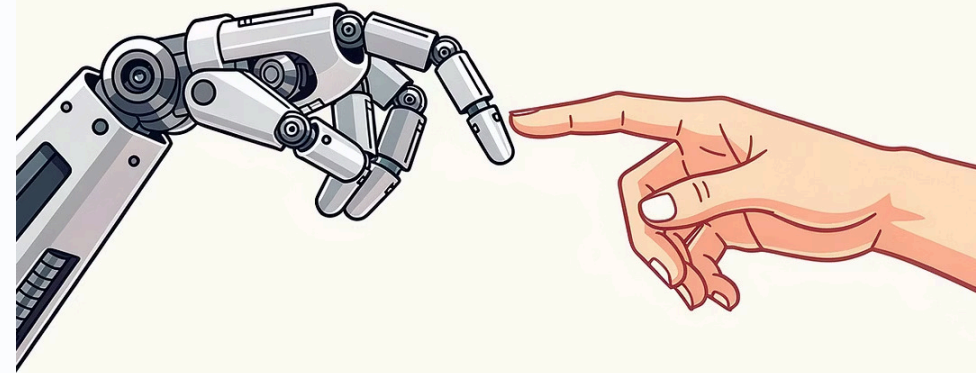
Session 10: Strengths & Limits of Generative AI

You've created with AI. You've seen it get things wrong. You've looked inside how it works. Now it's time to step back and ask the big questions:

What is AI
GREAT at?

Where does it
FAIL?

When should
you TRUST it?



Objectives:

By the end of this session, you'll be able to:

01

List what Generative AI is GREAT at

02

List what Generative AI
STRUGGLES with

03

Decide when to USE AI – and when
NOT TO

04

Understand the Human-in-the-Loop idea

05

Do your own quick AI Tool Analysis

Recap & Homework

Welcome back! Let's see your **Hallucination Hunt!**

Share Your Hunt

- What topic did you pick?
- Did AI get anything right?
- Did AI get anything wrong?
- Did anything feel sketchy?

Quick Check from Last Week

- What's an AI Hallucination?
- Confidence vs Accuracy — same or different?
- What's special about NotebookLM?





Warm-Up: The Tool Question

Imagine you have a **super-powered hammer**- it can pound nails 100x faster, swing itself, and even paints! Would you use it for *everything*?

Pound a nail?

Yes — that's exactly what it's for!

Cut bread?

No — it's a hammer, not a knife!

Hug your dog?

Definitely not!

① AI is the same. It's an amazing tool - but only for the right jobs.

AI Is GREAT At This Stuff

Generative AI has real superpowers. Let's name them:



Speed

Does in 10 seconds what takes you 1 hour.



Brainstorming

Endless ideas when you're stuck.



Drafts

Gives you a starting point to build from.



Explaining

Can re-explain something 5 different ways.



Languages

Translates in seconds across dozens of languages.



Creating

Stories, poems, images, songs — all from a prompt.

Strength Example #1

Brainstorming with AI

Your school is doing a "**Kindness Week**" and you need 10 creative ideas for activities. You're stuck. 😞

You ask ChatGPT: *"Give me 10 creative ideas for Kindness Week at a middle school."*

- ✅ In 5 seconds, you get a full list — anonymous compliment notes, kindness scavenger hunt, letter to a teacher you appreciate... and 7 more!

AI brainstormed. YOU chose. That's AI at its best.



Strength Example #2

Re-Explaining with AI

You're learning about **gravity** and the textbook just confuses you.

→ Ask: "Explain gravity like I'm 8."

"Imagine the Earth is a giant magnet for everything around it..."

→ Still confused? Ask again!

"Now explain it using a trampoline."

→ New explanation clicks!

"A bowling ball sinks into the trampoline, creating a dip. Roll a marble nearby..."

① One topic. Multiple explanations. Until ONE finally clicks. That's AI at its best.



But AI Has Real WEAKNESSES Too

AI is not perfect. It has real limits — pretty different from that superpower list!

-  **No Real Understanding**
It pattern-matches — it doesn't actually *think*.
-  **Can Hallucinate**
Confidently makes stuff up. (Remember Session 9!)
-  **Outdated Information**
Doesn't know about recent events.
-  **Can Repeat Biases**
Copies unfair patterns from its training data.
-  **No Common Sense**
Can miss things that are completely obvious to humans.
-  **Can't Tell True from False**
It doesn't fact-check itself.

Weakness Example #1

No Common Sense

Someone asks AI: "Can I fry an egg on the sun?"

🤖 AI says...

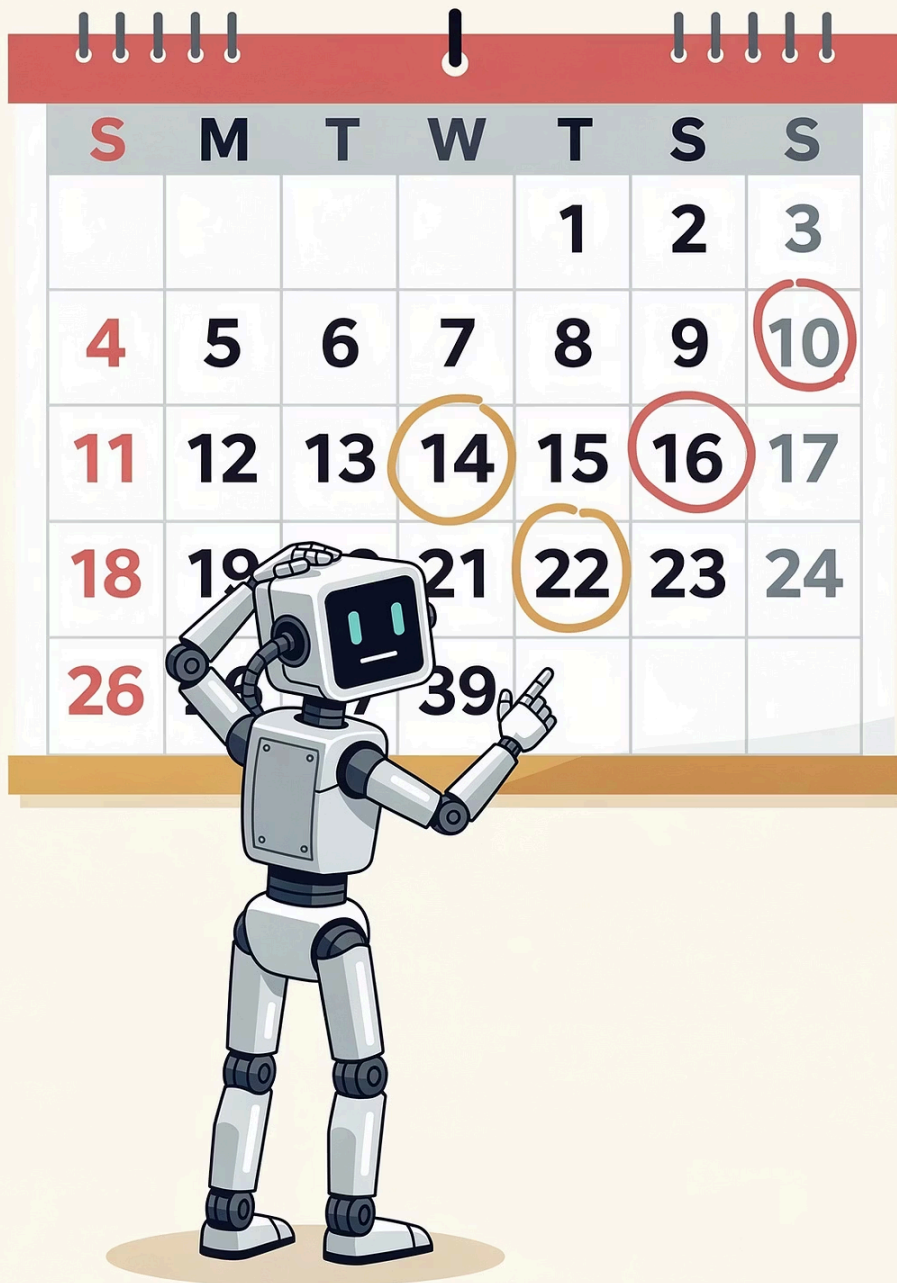
"The sun is too hot and far away, but *theoretically...*"
(*launches into physics*)

🧠 A 5-year-old says...

"That's silly!"

AI doesn't know it's silly. It doesn't have common sense — it has **patterns**.





Weakness Example #2: Outdated Info

You ask ChatGPT: "What did the President say in their speech *YESTERDAY*?"

😬 Option A

Makes something up that sounds real — a hallucination!

🙅 Option B

Says "I don't have access to events from yesterday."

🌐 Option C

Looks it up online — only if it has internet access.

⚠️ The problem: you can't always tell if AI is giving you **OLD info**, **REAL info**, or **MADE UP info**. For anything happening NOW — use Google or check the real source. Don't rely on AI alone.

Mini Activity 1: Strength or Weakness?

For each task below, decide: is this AI's **STRENGTH** or **WEAKNESS**?

Task	Strength or Weakness?
Brainstorming names for a new pet	?
Telling you who won the game last night	?
Writing a thank-you card draft	?
Telling you if a friend is actually upset with you right now	?
Explaining how clouds form 3 different ways	?
Recognizing if a joke is appropriate for grandma	?
Generating ideas for a science project	?
Predicting if it'll rain tomorrow at YOUR house	?

Mini Activity 1: Answers

Task	Answer	Why
Pet names	 Strength	Brainstorming = AI superpower
Last night's game	 Weakness	AI doesn't know recent events
Thank-you card draft	 Strength	Drafts = AI superpower
What time YOUR class period 1 starts today	 Weakness	AI doesn't know YOUR specific schedule
Explaining clouds 3 ways	 Strength	Re-explaining = AI superpower
If joke is grandma-appropriate	 Weakness	No common sense!
Science project ideas	 Strength	Brainstorming = AI superpower
Rain at YOUR house tomorrow	 Weakness	Outdated + can't access live data
 AI = Creative + General		 AI ≠ Specific + Current

When SHOULD You Use AI?

Use AI when you need a **starting point** — and YOUR job is to make it good.



Brainstorm Ideas

Need inspiration? AI can generate dozens of ideas fast.



Starting Draft

Get a rough draft to edit and make your own.



When You're Stuck

Hit a wall? AI can help you break through creative blocks.



Explain Differently

Want a topic explained in a new way? Ask AI to try again.



Quick Translation

Get a fast translation to understand or communicate across languages.



Create Something Fun

Poems, stories, art prompts — AI loves creative play.

When Should You NOT Use AI?

Don't fully trust AI for these situations — talk to humans, check real sources, or use specific tools instead.



Current News

AI's information can be old or incorrect. Use real news sources for what's happening now.



Medical Advice

AI is not a doctor. For health questions, always consult a real medical professional.



Serious or Risky Situations

Anything with real consequences needs real experts — not an AI guess.



Facts That Must Be Right

AI can sound confident and still be wrong. Always double-check specific facts.



Emergencies

Call a real person. In a crisis, AI is not the answer.



Info About Real People

AI can fabricate details about real individuals. Don't trust it without verification.

Mini Activity 2: Use AI or Not?

Real-life situations. For each one, decide: **AI YES** or **AI NO?**

Situation	Use AI?
Writing a birthday card for your friend	?
Figuring out if you're sick	?
Getting ideas for a story you're writing	?
Doing your math homework FOR you	?
Finding fun activities for a rainy day	?
Learning who won the soccer game last night	?
Coming up with a band name	?
Asking what your mom said this morning	?

Mini Activity 2: Answers

Situation	Answer	Why
Birthday card draft	✓ Use AI	Creative draft = great use
Are you sick?	✗ Don't	Ask a parent or doctor!
Story ideas	✓ Use AI	Brainstorming = perfect
Math homework FOR you	✗ Don't	If AI does your homework, YOU don't learn. Use AI to explain, not to replace you.
Rainy day activities	✓ Use AI	Brainstorming = great
Soccer game last night	✗ Don't	AI doesn't know recent events
Band name	✓ Use AI	Creative = perfect
What mom said this morning	✗ Don't	AI has no idea! Ask her!

 Pattern: AI helps you **THINK and CREATE** — but doesn't replace real people or fresh info.



The Big Idea: Human-in-the-Loop

The most important concept of this session:



AI Does the Work
Fast, tireless,
generates ideas and
drafts at incredible
speed.



YOU Stay in Charge
You make the
decisions, check the
facts, and apply
judgment.



Best Results
The combination of
AI's speed and your
wisdom wins every
time.

Human-in-the-Loop in Real Life

Planning a Class Assembly

Imagine you're planning a class assembly about kindness.

 AI Brainstorms 10 ideas in 5 seconds

 YOU do:

- **Pick What Makes Sense**
Choose ideas that actually fit **your** school's culture and students.
- **Check the Budget**
Realize some ideas cost too much and rule them out — AI doesn't know your limits.
- **Choose the 3 Best**
Apply judgment, creativity, and context to narrow down to what will actually work.
- **Get Teacher Approval**
Bring in the humans who have authority and responsibility for the outcome.
- **Plan the Actual Day**
Execute, adapt, and lead — the parts that require a real human presence.

✓ AI assisted. YOU directed. That's Human-in-the-Loop. You stayed in charge the whole time.





The Pilot Story

Airplanes have **autopilot** - AI that helps fly the plane. But here's the thing: there's **ALWAYS** a real pilot in the cockpit.

Sets the Destination

The pilot decides where the plane is going - autopilot never chooses the course on its own.

Watches What Autopilot Does

Constant monitoring ensures the automated system is performing correctly at every stage.

Takes Over If Anything Goes Wrong

When conditions change or something unexpected happens, the pilot steps in immediately.

Is Responsible for the Flight

Accountability never transfers to the machine. The pilot owns the outcome, always.

You = the pilot. AI = the autopilot. You're always in charge.

Hands-On Activity: Your AI Tool Analysis

Time to pull it all together. Pick **ONE AI tool** you've used in this course:



ChatGPT



Gemini



NotebookLM



Google Storybook

You're going to do your own **AI Tool Analysis** — a personal guide to using AI wisely.

AI Tool Analysis Template

Pick one tool, then answer these questions together:

01

Which tool did you pick?

02

What is it GREAT at?
(1-2 examples)

03

What does it STRUGGLE with?
(1-2 examples)

04

Share one good prompt you've used

05

What can't this tool do well?

06

When would you USE this tool?

07

When would you NOT use it?

08

Your personal rule for using AI?



Share Your Analysis!

Walk me through what you found:

- What surprised you?
- What's the tool's biggest strength?
- What's its biggest weakness?
- What's YOUR personal rule for using AI?

“

"I'll always double-check facts before sharing them."

”

“

"I'll use AI for brainstorming, but write my own first drafts."

”

“

"I'll never use AI for things only I can decide."

”

“

"If I can't explain it in my own words after AI helps, I haven't really learned it."

”

What We Learned in this Session

AI is GREAT at...

- Speed & efficiency
- Brainstorming ideas
- Writing drafts
- Explaining concepts
- Translating content

AI is WEAK at...

- Common sense reasoning
- Current information
- Specific verified facts
- Your personal context

The Pattern

AI = **creative + general**

AI ≠ **specific + current**

Human-in-the-Loop

AI helps. **YOU decide.** Like a pilot - always in charge.

Your AI Tool Analysis

Your personal AI guide - built by you, for you.

👉 You're now a **smart AI user**. That's not common.

Your Homework: AI Decisions Diary

For **3 days**, every time you **use** or **could have used** AI - write it down. Just 5 minutes a day.

Day	Situation	Did you use AI?	Was it the right call?	What did you learn?
Day 1	Stuck on a story idea	Used ChatGPT	Yes great for ideas!	AI is great when I'm blocked
Day 2				
Day 3				

  **Goal:** Notice when AI helps - and when it would be better to **NOT** use it. Just 5 minutes a day for 3 days.





Next Time: Session 11 - The Art of Prompting

You've used AI a lot now. You've seen what makes a good prompt vs a bad prompt. Next session, we make it official:



How to write great prompts every time



The 5-part prompt framework



How to fix a prompt that didn't work



You'll start your own Prompt Library!



Bring your AI Decisions Diary! We'll use it in the next session.